

1. A particle of mass 10 g moves in a straight line with retardation  $2x$ , where  $x$  is the displacement in SI units. Its loss of kinetic energy for above displacement is  $\left(\frac{10}{x}\right)^{-n}$  J. The value of  $n$  will be \_\_\_\_\_.

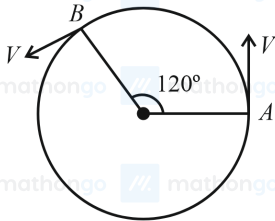
[2023 (06 Apr Shift 1)]

2. A particle starts with an initial velocity of  $10.0 \text{ ms}^{-1}$  along  $x$ -direction and accelerates uniformly at the rate of  $2.0 \text{ m s}^{-2}$ . The time taken by the particle to reach the velocity of  $60.0 \text{ m s}^{-1}$  is \_\_\_\_\_.

[2023 (06 Apr Shift 2)]

- (1) 25 s
- (2) 3 s
- (3) 6 s
- (4) 30 s

3. As shown in the figure, a particle is moving with constant speed  $\pi \text{ m s}^{-1}$ . Considering its motion from  $A$  to  $B$ , the magnitude of the average velocity is:



[2023 (06 Apr Shift 2)]

- (1)  $\sqrt{3} \text{ m s}^{-1}$
- (2)  $\pi \text{ m s}^{-1}$
- (3)  $1.5\sqrt{3} \text{ m s}^{-1}$
- (4)  $2\sqrt{3} \text{ m s}^{-1}$

4. Given below are two statements:

**Statement I:** Area under velocity-time graph gives the distance travelled by the body in a given time.

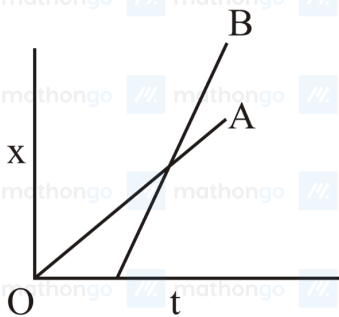
**Statement II:** Area under acceleration-time graph is equal to the change in velocity in the given time.

In the light of given statements, choose the correct answer from the options given below.

[2023 (08 Apr Shift 2)]

- (1) Both Statement I and Statement II are true
- (2) Both Statement I and Statement II are false
- (3) Statement I is correct but Statement II is false
- (4) Statement I is incorrect but Statement II is true

5. The position-time graphs for two students  $A$  and  $B$  returning from the school to their homes are shown in figure.



- (A)  $A$  lives closer to the school
- (B)  $B$  lives closer to the school
- (C)  $A$  takes lesser time to reach home
- (D)  $A$  travels faster than  $B$
- (E)  $B$  travels faster than  $A$

Choose the correct answer from the options given below

[2023 (10 Apr Shift 1)]

- (1) (A), (C) and (D) only
- (2) (A), (C) and (E) only
- (3) (B) and (E) only
- (4) (A) and (E) only

6. A person travels  $x$  distance with velocity  $v_1$  and then  $x$  distance with velocity  $v_2$  in the same direction. The average velocity of the person is  $v$ , then the relation between  $v$ ,  $v_1$  and  $v_2$  will be

[2023 (10 Apr Shift 2)]

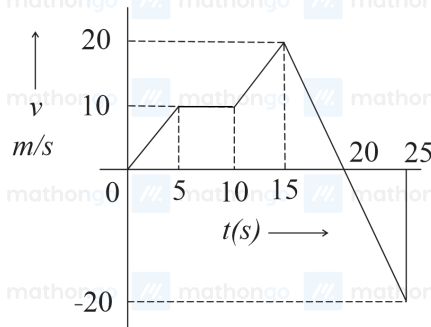
(1)  $v = \frac{v_1 + v_2}{2}$

(2)  $\frac{1}{v} = \frac{1}{v_1} + \frac{1}{v_2}$

(3)  $v = v_1 + v_2$

(4)  $\frac{2}{v} = \frac{1}{v_1} + \frac{1}{v_2}$

7. From the  $v - t$  graph shown, the ratio of distance to displacement in 25 s of motion is:



[2023 (11 Apr Shift 1)]

(1) 1

(2)  $\frac{1}{2}$

(3)  $\frac{5}{3}$

(4)  $\frac{3}{5}$

8. A ball is thrown vertically upward with an initial velocity of  $150 \text{ m s}^{-1}$ . The ratio of velocity after 3 s and 5 s is  $\frac{x+1}{x}$ . The value of  $x$  is \_\_\_\_\_. {take,  $g = 10 \text{ m s}^{-2}$ }

[2023 (12 Apr Shift 1)]

(1) 10

(2) -5

(3) 6

(4) 5

9. Given below are two statements:

**Statement I :** A truck and a car moving with same kinetic energy are brought to rest by applying breaks which provide equal retarding forces. Both come to rest in equal distance.

**Statement II :** A car moving towards east takes a turn and moves towards north, the speed remains unchanged. The acceleration of the car is zero.

In the light of given statements, choose the most appropriate answer from the options given below

[2023 (12 Apr Shift 1)]

(1) Statement I is correct but statement II is incorrect

(2) Statement I is incorrect but statement II is correct

(3) Both statement I and Statement II are correct

(4) Both statement I and statement II are incorrect

10. Two trains A and B of length  $l$  and  $4l$  are travelling into a tunnel of length  $L$  in parallel tracks from opposite directions with velocities  $108 \text{ km h}^{-1}$  and  $72 \text{ km h}^{-1}$ , respectively. If train A take 35 s less time than train B to cross the tunnel then, length  $L$  of tunnel is:

(Given  $L = 60 l$ )

[2023 (13 Apr Shift 1)]

(1) 1200 m

(2) 900 m

(3) 1800 m

(4) 2700 m

11. The distance travelled by an object in time  $t$  is given by  $s = (2.5)t^2$ . The instantaneous speed of the object at  $t = 5 \text{ s}$  will be :

[2023 (13 Apr Shift 2)]

(1)  $25 \text{ m s}^{-1}$

(2)  $5 \text{ m s}^{-1}$

(3)  $62.5 \text{ m s}^{-1}$

(4)  $12.5 \text{ m s}^{-1}$

12. A passenger sitting in a train A moving at  $90 \text{ km h}^{-1}$  observes another train B moving in the opposite direction for 8 s. If the velocity of the train B is  $54 \text{ km h}^{-1}$ , then length of train B is:

[2023 (13 Apr Shift 2)]

- (1) 120 m  
(2) 320 m  
(3) 80 m  
(4) 200 m

13. The position of a particle related to time is given by  $x = (5t^2 - 4t + 5) \text{ m}$ . The magnitude of velocity of the particle at  $t = 2 \text{ s}$  will be :

[2023 (15 Apr Shift 1)]

- (1)  $06 \text{ m s}^{-1}$   
(2)  $14 \text{ m s}^{-1}$   
(3)  $10 \text{ m s}^{-1}$   
(4)  $16 \text{ m s}^{-1}$

## ANSWER KEYS

1. (2)      2. (1)      3. (3)      4. (4)      5. (4)      6. (4)      7. (3)      8. (4)  
9. (1)      10. (3)      11. (1)      12. (2)      13. (4)

1. (2)

Acceleration can be written as,  $a = v \frac{dv}{dx}$ .

Given:  $a = -2x$  (-ve sign is due to retardation)

Therefore,

$$v \frac{dv}{dx} = -2x$$

$$\Rightarrow v dv = (-2x) dx$$

Integrating both sides, we get

$$\frac{1}{2} (v_f^2 - v_i^2) = -2 \left( \frac{x^2}{2} \right) = -x^2$$

$$\Rightarrow \left| \frac{1}{2} m (v_f^2 - v_i^2) \right| = m x^2$$

$$\Rightarrow |\Delta KE| = 0.01 x^2 = \left( \frac{10}{x} \right)^{-2}$$

Hence,  $n = 2$ .

2. (1)

The given data is

$$v = 60 \text{ m s}^{-1}$$

$$u = 10 \text{ m s}^{-1}$$

$$a = 2 \text{ m s}^{-2}$$

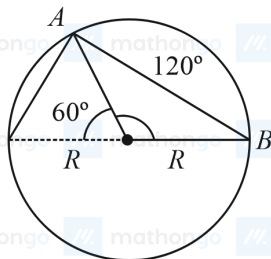
Using the equation  $v = u + at$ , we get

$$60 = 10 + 2t$$

$$\Rightarrow t = \frac{(60-10)}{2}$$

$$\Rightarrow t = 25 \text{ s}$$

3. (3)



The given data is

$$v = \pi \text{ m s}^{-1}$$

By using trigonometric ratios in the triangle shown above the displacement can be found out by

$$\sin 60^\circ = \frac{AB}{2R}$$

$$\Rightarrow AB = R\sqrt{3}$$

The formula for distance is

$$d = vt$$

$$\Rightarrow t = \frac{d}{v}$$

$$\Rightarrow t = \frac{120^\circ}{360^\circ} \times \frac{2\pi R}{\pi} = \frac{2R}{3}$$

So, the average velocity is

$$v_{\text{avg}} = \frac{\sqrt{3}R}{\frac{2R}{3}} = \frac{3\sqrt{3}}{2}$$

$$= 1.5\sqrt{3} \text{ m s}^{-1}$$

4. (4)

The definition of displacement is given by the product of velocity and time.

$\int v dt = \Delta x$  = displacement of body in given time.

Distance is a scalar quantity and displacement is a vector quantity defined by the shortest distance between the initial and the final position.

Acceleration is defined by the rate of change of velocity.

$\int a dt = \Delta v$  = change in velocity in given time.

Thus, the area under the velocity time graph gives the displacement of the body and the area under the acceleration time graph gives the change in velocity of the body.

5. (4)

From the position time graph the velocity is found out by the slope of the line. The slope is given by  $\frac{x_2 - x_1}{t_2 - t_1}$ .

The slope of B is greater than A. Hence, the velocity of B is greater than A. As can be seen from the graph  $X_A < X_B$ . So A lives closer to the school and  $V_B > V_A$ . Hence, this is the correct option.

6. (4)

The time ( $t_1$ ) taken by the person to travel distance  $x$  with velocity  $v_1$  is given by

$$t_1 = \frac{x}{v_1} \dots (1)$$

The time ( $t_2$ ) taken by the person to travel distance  $x$  with velocity  $v_2$  is given by

$$t_2 = \frac{x}{v_2} \dots (2)$$

The formula to calculate the average speed ( $v$ ) of the person is given by

$$v = \frac{2x}{t_1 + t_2} \dots (3)$$

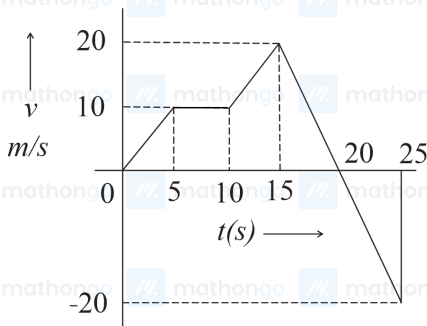
Substitute the expressions from equations (1) and (2) into equation (3) to obtain the required relation.

$$v = \frac{2x}{\left(\frac{x}{v_1} + \frac{x}{v_2}\right)}$$

$$\Rightarrow \frac{x}{v_1} + \frac{x}{v_2} = \frac{2x}{v}$$

$$\Rightarrow \frac{2}{v} = \frac{1}{v_1} + \frac{1}{v_2}$$

7. (3)



The area under the velocity time graph gives the displacement. The area for  $t = 0$  s to  $t = 25$  s is

$$(Area) = \left(\frac{1}{2} \times 5 \times 10\right) + (5 \times 10) + \left(\frac{1}{2} \times 30 \times 5\right) + \left(\frac{1}{2} \times 20 \times 5\right) - \left(\frac{1}{2} \times 20 \times 5\right)$$

$$\Rightarrow Area = 25 + 50 + 75 + 50 - 50 = 150 \text{ m}$$

Hence, the displacement = 150 m

The distance is defined as the path taken by the object to travel from initial to final position. Also note that displacement is a vector quantity, but distance is a scalar quantity.

Distance

$$= \left(\frac{1}{2} \times 5 \times 10\right) + (5 \times 10) + \left(\frac{1}{2} \times 30 \times 5\right) + \left(\frac{1}{2} \times 20 \times 5\right) + \left(\frac{1}{2} \times 20 \times 5\right)$$

$$= (200 + 50) = 250 \text{ m}$$

Therefore, required ratio

$$\frac{\text{distance}}{\text{displacement}} = \frac{250}{150} = \left(\frac{5}{3}\right)$$

8. (4)

The formula to calculate the velocity of the ball after 3 s is given by

$$v_3 = u - 3g \dots (1)$$

The formula to calculate the velocity of the ball after 5 s is given by

$$v_5 = u - 5g \dots (2)$$

Divide equation (1) by equation (2) to obtain the required ratio.

$$\frac{v_3}{v_5} = \frac{u-3g}{u-5g} \dots (3)$$

Substitute the values of the known parameters into equation (3) to calculate the required ratio.

$$\begin{aligned} \frac{v_3}{v_5} &= \frac{150-30}{150-50} \\ &= \frac{120}{100} \\ &= \frac{6}{5} \\ &= \frac{5+1}{5} \end{aligned}$$

Comparing the above equation with the given expression, it can be written that  $x = 5$ .

9. (1)

The work done in stopping the car and the truck can be written as

$$W = Fx$$

$$= \Delta KE$$

As both the car and the truck are moving initially with the same kinetic energy, after being stopped, they will have the same change of kinetic energy.

Also, they are stopped with the same applied force.

Hence, from the above equation, it can be written that they have the same displacement, i.e.,  $x_1 = x_2$ .

Acceleration is a vector quantity. As the direction of motion of the car is different in the two given situations, the acceleration of the car will not be the same.

10. (3)

The time taken for A

$$\Delta t_A = \frac{l+L}{V_A}$$

The time taken for B

$$\Delta t_B = \frac{4l+L}{V_B}$$

It is given that B takes more time than A. So,

$$\Rightarrow \frac{4l+L}{20} - \frac{l+L}{30} = 35$$

$$\Rightarrow \frac{64l}{20} - \frac{61l}{30} = 35$$

$$\Rightarrow l = \frac{35 \times 600}{700} = 30 \text{ m}$$

Therefore,

$$L = 1800 \text{ m.}$$

11. (1)

It is given that

$$s = 2.5t^2$$

Differentiating,

$$v = \frac{ds}{dt} = \frac{d(2.5t^2)}{dt} = 5t$$

$$\text{At } t = 5 \text{ s, } v = 25 \text{ m s}^{-1}$$

12. (2)

Let the velocity of A be  $v_A$  and the velocity of B be  $v_B$ .

The relative velocity is

$$\vec{v}_{BA} = \vec{v}_B - \vec{v}_A$$

$$\Rightarrow v_{BA} = 54 - (-90) = 144 \text{ km h}^{-1}$$

$$\Rightarrow v_{BA} = 40 \text{ m s}^{-1}$$

$$\text{Time} = \frac{\text{length}}{v_{BA}}$$

$$\Rightarrow \text{length} = 40 \text{ m s}^{-1} \times 8 \text{ s} = 320 \text{ m}$$

13. (4) mathongo mathongo mathongo mathongo mathongo mathongo mathongo mathongo mathongo mathongo n

It is given that

$$x = 5t^2 - 4t + 5$$

The velocity is given by

$$v = \frac{dx}{dt} = \frac{d(5t^2 - 4t + 5)}{dt} = 10t - 4$$

At  $t = 2$  s,

$$v = 10(2) - 4 = 16 \text{ m s}^{-1}$$

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